

## ST STEPHEN, NB, Canada

WMO: 716070

Lat: 45.2103N

Lon: 67.2528W

Elev: 86

StdP: 14.65

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.1       | -4.1 | -19.2                         | 1.9 | -5.5 | -14.4 | 2.5 | -1.7 | 19.7                  | 29.6 | 17.7 | 24.1 | 2.5                   | 290  | 0.540 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.7                   | 85.6            | 69.6 | 82.3 | 68.2 | 79.3 | 66.3 | 72.3                | 81.9 | 70.5 | 79.3 | 68.6 | 76.4 | 7.1                  | 250  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.1                            | 107.3 | 77.0 | 67.4 | 101.1 | 74.6 | 65.7 | 95.1 | 72.6 | 36.0          | 81.7 | 34.4 | 79.4 | 32.9 | 76.5 | 81.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 16.6          | 14.4 | 12.7 | DB                         | -16.8 | 91.5               | 6.2 | 2.4                                                | -21.2 | 93.2       | -24.8 | 94.7       | -28.3 | 96.0       | -32.7 | 97.8 | (4) |
| (5)               |      |      | WB                         | -17.5 | 75.0               | 5.3 | 1.7                                                | -21.2 | 76.3       | -24.3 | 77.3       | -27.3 | 78.3       | -31.1 | 79.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 43.5        | 18.1  | 20.6  | 29.5  | 41.1  | 51.7  | 60.5  | 66.8  | 65.6  | 58.1  | 47.0  | 36.5  | 25.4  | (6)  |  |
|                                                                        | DBStd       | 18.57       | 11.89 | 10.96 | 9.45  | 6.66  | 6.04  | 5.99  | 4.75  | 4.94  | 6.91  | 7.17  | 8.62  | 10.17 | (7)  |  |
|                                                                        | HDD50       | 4104        | 989   | 822   | 636   | 277   | 49    | 2     | 0     | 0     | 9     | 146   | 413   | 761   | (8)  |  |
|                                                                        | HDD65       | 8046        | 1454  | 1242  | 1100  | 719   | 414   | 160   | 36    | 53    | 226   | 560   | 856   | 1226  | (9)  |  |
|                                                                        | CDD50       | 1743        | 0     | 0     | 1     | 9     | 102   | 317   | 520   | 483   | 253   | 51    | 7     | 0     | (10) |  |
|                                                                        | CDD65       | 208         | 0     | 0     | 0     | 0     | 2     | 25    | 91    | 71    | 19    | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 2290        | 0     | 0     | 4     | 4     | 77    | 376   | 919   | 693   | 209   | 8     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 553         | 0     | 0     | 0     | 1     | 15    | 98    | 223   | 160   | 56    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.8         | 5.3   | 5.7   | 6.1   | 5.9   | 5.1   | 4.3   | 3.8   | 3.5   | 3.8   | 4.6   | 4.9   | 5.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 49.7        | 3.9   | 3.4   | 4.3   | 3.8   | 3.9   | 3.9   | 3.7   | 3.3   | 4.1   | 5.4   | 4.8   | 5.3   | (15) |  |
|                                                                        | PrecMax     | 70.9        | 8.1   | 7.4   | 7.5   | 6.9   | 7.9   | 7.6   | 11.1  | 7.6   | 10.2  | 12.2  | 7.6   | 11.8  | (16) |  |
|                                                                        | PrecMin     | 30.5        | 1.6   | 0.9   | 1.3   | 0.9   | 1.0   | 1.8   | 0.9   | 0.4   | 1.7   | 1.9   | 2.4   | 2.2   | (17) |  |
|                                                                        | PrecStd     | 9.3         | 1.5   | 1.4   | 1.6   | 1.5   | 1.7   | 1.7   | 2.2   | 1.9   | 1.9   | 2.6   | 1.7   | 2.3   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 53.0  | 50.8  | 61.7  | 69.9  | 81.4  | 88.1  | 89.7  | 88.7  | 87.2  | 74.1  | 62.7  | 55.6  | (19) |  |
|                                                                        |             | MCWB        | 51.6  | 46.6  | 49.7  | 53.7  | 63.5  | 69.6  | 71.8  | 71.0  | 71.7  | 62.2  | 58.5  | 54.0  | (20) |  |
|                                                                        | 2%          | DB          | 46.9  | 44.2  | 51.0  | 62.7  | 75.1  | 82.8  | 85.3  | 84.3  | 79.7  | 67.5  | 58.5  | 49.4  | (21) |  |
|                                                                        |             | MCWB        | 45.3  | 39.8  | 42.8  | 49.6  | 59.3  | 67.4  | 69.9  | 69.4  | 66.9  | 58.7  | 55.4  | 47.8  | (22) |  |
|                                                                        | 5%          | DB          | 40.5  | 40.0  | 46.8  | 57.8  | 70.0  | 78.2  | 82.2  | 80.9  | 74.9  | 63.3  | 55.0  | 44.7  | (23) |  |
|                                                                        |             | MCWB        | 38.2  | 36.6  | 40.9  | 47.7  | 56.4  | 64.4  | 68.4  | 67.9  | 64.6  | 57.6  | 51.7  | 42.7  | (24) |  |
|                                                                        | 10%         | DB          | 35.1  | 36.8  | 42.8  | 53.9  | 65.6  | 74.3  | 79.2  | 77.7  | 70.9  | 60.1  | 50.8  | 40.2  | (25) |  |
|                                                                        |             | MCWB        | 33.2  | 33.9  | 37.7  | 45.7  | 54.0  | 61.9  | 66.7  | 65.8  | 62.1  | 55.3  | 47.8  | 37.9  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 52.0  | 47.6  | 52.5  | 57.8  | 66.5  | 72.8  | 74.6  | 74.3  | 72.5  | 64.9  | 60.5  | 54.6  | (27) |  |
|                                                                        |             | MCDB        | 52.8  | 49.7  | 58.5  | 64.2  | 76.7  | 84.8  | 83.9  | 83.5  | 83.7  | 68.4  | 61.7  | 55.5  | (28) |  |
|                                                                        | 2%          | WB          | 45.4  | 41.2  | 46.1  | 53.7  | 61.7  | 69.5  | 72.4  | 71.9  | 69.2  | 61.8  | 56.3  | 48.3  | (29) |  |
|                                                                        |             | MCDB        | 46.4  | 43.1  | 48.8  | 58.2  | 71.0  | 79.2  | 81.8  | 80.5  | 76.3  | 65.0  | 58.0  | 49.1  | (30) |  |
|                                                                        | 5%          | WB          | 38.4  | 37.3  | 41.7  | 49.9  | 58.6  | 66.7  | 70.6  | 69.9  | 66.6  | 59.1  | 52.3  | 43.2  | (31) |  |
|                                                                        |             | MCDB        | 39.8  | 39.5  | 45.3  | 55.2  | 66.4  | 75.0  | 79.5  | 78.4  | 72.0  | 62.6  | 54.1  | 44.5  | (32) |  |
|                                                                        | 10%         | WB          | 33.6  | 34.1  | 38.4  | 46.6  | 55.8  | 64.0  | 68.7  | 67.9  | 64.2  | 55.8  | 48.0  | 38.2  | (33) |  |
|                                                                        |             | MCDB        | 35.1  | 36.3  | 42.1  | 52.5  | 62.6  | 71.9  | 76.5  | 74.8  | 69.2  | 59.1  | 50.3  | 39.9  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 19.9  | 20.5  | 19.0  | 19.7  | 22.0  | 21.8  | 21.7  | 22.1  | 22.4  | 20.0  | 16.4  | 17.0  | (35) |  |
|                                                                        |             | MCDBR       | 24.4  | 21.7  | 23.7  | 29.1  | 32.2  | 30.2  | 27.1  | 27.1  | 27.4  | 25.5  | 21.1  | 21.1  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 23.8  | 18.3  | 16.9  | 17.9  | 18.2  | 16.6  | 13.9  | 14.1  | 16.2  | 17.7  | 18.0  | 20.0  | (37) |  |
|                                                                        |             | MCWBR       | 24.6  | 19.7  | 21.0  | 23.2  | 26.2  | 25.8  | 23.1  | 22.8  | 22.1  | 21.2  | 18.4  | 20.2  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.279 | 0.291 | 0.320 | 0.348 | 0.369 | 0.408 | 0.425 | 0.397 | 0.362 | 0.328 | 0.308 | 0.291 | (40) |  |
|                                                                        |             | taud        | 2.478 | 2.447 | 2.455 | 2.429 | 2.414 | 2.319 | 2.288 | 2.368 | 2.436 | 2.534 | 2.545 | 2.489 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 269   | 286   | 291   | 290   | 285   | 273   | 267   | 271   | 272   | 269   | 256   | 251   | (42) |  |
|                                                                        |             | Edh at Noon | 22    | 27    | 30    | 34    | 36    | 40    | 41    | 36    | 31    | 24    | 20    | 19    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 481   | 755   | 1125  | 1389  | 1587  | 1721  | 1768  | 1571  | 1282  | 808   | 480   | 368   | (44) |  |
|                                                                        | RadStd      | RadStd      | 45    | 62    | 75    | 131   | 189   | 143   | 137   | 100   | 87    | 77    | 50    | 36    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | +148  | N/A   |       |
| (47) Regional (28 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | +0.72 | N/A | N/A         | +97   | N/A   |       |

Nomenclature: See separate page